

		$n = \frac{3(\sin 30^\circ)}{\sin 15^\circ} = 1.21$ Hence the highest order which can be viewed on the screen is the 4th order.
20	A	Electric potential = $\frac{Q}{4\pi\epsilon_0(\frac{1}{3}r)} + \frac{2Q}{4\pi\epsilon_0(\frac{2}{3}r)} = \frac{3Q}{2\pi\epsilon_0 r}$ Electric field strength = $\frac{Q}{4\pi\epsilon_0 (\frac{1}{3}r)^2} - \frac{2Q}{4\pi\epsilon_0 (\frac{2}{3}r)^2} = \frac{9Q}{8\pi\epsilon_0 r^2}$
21	D	Using Fleming's Left Hand Rule, the magnetic force on the protons is upwards. Hence the